

Ryan Ye

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Education

Cornell University

Ithaca, NY

B.S. in Computer Science, College of Engineering, GPA 4.02

Expected May 2028

- **Coursework:** Deep Learning, Computer Vision, Machine Learning, Computer System Organization (C), Algorithms, Data Structures, Functional Programming (OCaml), Discrete Math, Linear Algebra, Differential Equations, Multivariable Calculus
- **Honors:** BURE Fellow, CIDA Grant Recipient, AIMI Junior Affiliate, 4-time AIME Qualifier, USACO Silver Division

Technical Skills

- **Languages:** Python, Java, C/C++, OCaml, JavaScript (Node.js, Express.js, React), HTML/CSS
- **Libraries:** PyTorch, JAX, Hugging Face Transformers and PEFT, NumPy, pandas, Matplotlib
- **Tools:** Git, SSH, SLURM/HPC, CUDA, Jupyter, VS Code

Research Experience

Undergraduate Researcher, Sun Lab at Cornell University

May 2025 – Present

PI: Jennifer Sun. AI for science: scientific discovery agents and computer vision for animal behavior

- **Statistical validation for automated hypothesis discovery (AIMI Junior Affiliate project), Jun 2026 – Present.** Implemented a separate statistical check on what AI discovery agents report: one pre-set test per finding on held-out data, then a standard false discovery rate correction over everything the agent submitted
- Ran several agent systems on test datasets with known true relationships, where the check held false findings under the target rate for every agent and the agents' own reports did not; showed in stress tests that its error control can fail when held-out results are fed back to the agent
- **Multi-lab video behavior recognition (MABe), Feb 2026 – Present.** Beat the frozen-backbone baseline on 9 of 15 lab datasets (13 of 15 on at least one split) by LoRA fine-tuning a frozen video foundation model with a self-authored kinematic auxiliary loss; on one lab, video alone beat the pose-tracking ceiling
- Built the 8-stage SLURM pipeline with 5-fold cross-validation and ensemble threshold calibration across all 15 datasets, staging up to 1.9 TB of video per lab on node-local NVMe; brought it up on Blackwell GPUs (CUDA 12.8)
- **Calf behavior monitoring (BURE Fellow, CIDA grant), May 2025 – Present.** Mean posture-classification F1 rose from 0.928 with 300 training images to 0.957 with 1,000 (24 runs per setting, DINOv2 vs. DINOv3 ensembles); fine-tuned YOLO on calf data I annotated in CVAT. Poster presented at ADSA 2026; manuscript in preparation

Projects

Worship Rig

Sep 2026

Personal project. Electron, JavaScript, Web Audio

Site · GitHub

- Built and shipped a Mac app that turns a MIDI keyboard into a keys rig for worship bands: a Web Audio engine with 43 instruments and a key drone, a GarageBand sample converter, and 1,000+ automated tests

DoRA Re-Implementation and Evaluation

Spring 2026

CS 4782 Deep Learning final project, 4-person team. PyTorch, Hugging Face PEFT

GitHub

- Team re-implementation of DoRA, tested against LoRA on Qwen2.5-3B (AMC12, MATH); ran all HPC experiments
- Built the evaluation harness (logit-probe and two-pass scoring separating what a model knows from what it generates), the data pipeline with a leakage filter, and the analysis and figures; diagnosed a critical bug in the DoRA layer

Traveling Salesman: Conversational Flight Search

Mar 2026

Cornell AI Hackathon, frontend. React, d3-geo

GitHub

- Built most of the React frontend: the trip planner, ranked flight plans, and routes on a 2D map and 3D globe

Sisyphus: AI Productivity App

Nov 2025

Claude Builders Club Hackathon, 4-person team. Node.js, Express.js, Claude API

GitHub

- One of 5 winners of 38 projects (Fun Award): checks photo check-ins with Claude; worked on the frontend and AI validation

Publications

- **R. M. Ye**, M. Madureira Ferreira, J. J. Sun, F. A. Leal Yepes. “Automating pose detection in calves: a computer vision framework for non-invasive behavioral monitoring.” Poster, ADSA Annual Meeting, 2026
- M. Madureira Ferreira, K. M. Dadallage, T. Ward, H. McCray, **R. M. Ye**, J. J. Sun, F. A. Leal Yepes. “A smarter lung ultrasonography solution for lung consolidation classification in dairy calves using a deep learning approach.” *Journal of Dairy Science*, 2026. Supporting author
- **R. Ye**, Y. Chen, K. A. Kelly. “The effects of firm performance on CEO compensation and CEO pay ratio before and during COVID-19.” *Research in Economics* 77(4), 453–458, 2023. First author

Leadership

Basic Coding, Co-Founder & President. Free coding classes for 300+ students in 8 states

Aug 2020 – Jun 2024